

■ Secure File Storage with Encryption

This project is a modern and secure application for encrypting and decrypting files. It combines a simple **Graphical User Interface** built with **Tkinter** and strong **cryptography standards** to provide a safe way to store sensitive data.

■ Features

- ✓■ Password-based file encryption and decryption
- ✓■ PBKDF2-HMAC-SHA256 key derivation with random salt
- ✓■ Fernet (AES + HMAC) for authenticated encryption
- ✓■ Encrypted files saved in **Encrypted_Files/**
- ✓■ Decrypted files saved in **Decrypted_Files/**
- ✓■ User-friendly Tkinter interface

■■ Requirements

- Python 3.8 or above
- `pip install cryptography`
- Tkinter (default with Python on Windows, install separately on Linux/macOS)

■ How to Use

1. Run the program: `python secure_storage.py`
2. Enter a password in the input box.
3. To **Encrypt**: Click 'Encrypt File' → select file → file saved in **Encrypted_Files/**.
4. To **Decrypt**: Click 'Decrypt File' → select .enc file → enter same password → file saved in **Decrypted_Files/**.
5. Wrong password will result in an error message.

■ Security Notes

- The password is **never stored**; you must remember it.
- Losing the password makes the file unrecoverable.
- Always use a **strong password** (12+ characters, mix of letters, numbers, symbols).
- The .enc file only contains *salt* (public) and *ciphertext* (encrypted). Without the password, decryption is impossible.

■ With this tool, you can secure your files confidently and protect sensitive data from unauthorized access.